Medical Visual Question Answering for Chest X-rays

A knowledge-grounded approach with CheXagent-2-3b:
literature review, baseline tests and retrieval-augmented design

PROJECT REPORT

Supervisor	[Supervisor name, designation]
Submitted by	[Team member 1, roll number] [Team member 2, roll number] [Team member 3, roll number]
Department	[Department / programme]
Institution	[Institution name]
Date	[Month Year]

Names, institution and date are placeholders and will be replaced with the official title-page format.

Abstract

Visual question answering (VQA) lets a person ask a free-text question about a medical image and receive a free-text answer. This project studies VQA on chest X-rays with a small open model, CheXagent-2-3b, that runs on a free Colab T4 GPU. We reviewed six papers, kept three that shaped the plan (CheXagent, RULE, Med-Flamingo), and built a pre-registered evaluation: the rules, test sets and success conditions were written down before any model run. Zero-shot tests on 219 baseline questions, 130 knowledge questions, 140 left/right questions and a set of mirror-image pairs gave an overall accuracy of 58%, with weak left/right and knowledge scores. Disease naming on 130 images then produced the same answer for every image, which led us to the central finding of the work so far: **in every run to date the image never reached the model**, because of a software-version incompatibility. The earlier numbers are therefore a text-only control, not a measurement of the model's perception. We fixed the problem, added an automatic image check to every notebook, and corrected all earlier documents. In parallel we designed a knowledge-grounded retrieval pipeline (name the disease, then retrieve from MedlinePlus and medical textbooks) and measured, without any model, how often the corpus holds the gold answer: 29% of gold items on held-out questions with the true disease, 17% with a wrong disease and 3% from the question alone. No result with the image visible exists yet; those runs are the next step. The work makes no clinical-validity claim.

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1 Introduction

1.1 Motivation

Chest X-rays are among the most common medical images, and reading them takes training. A system that can answer plain questions about an X-ray ("Is the heart enlarged?", "Which side is the fluid on?", "What causes this disease?") could one day support teaching and report drafting. Recent vision-language models can produce such answers, but how reliable they are, and why, is not well understood. This project is a careful, small-scale investigation of that question, not a clinical tool.

1.2 Objectives

1. Explain what chest X-rays show and what medical VQA asks of a model.
2. Review the literature and keep only the findings that change what we do.
3. Measure how a ready-made chest X-ray model, CheXagent-2-3b, answers questions without any extra training.
4. Design and test a knowledge-grounded retrieval extension, and decide beforehand how success will be judged.
5. Report honestly, including results that went against our expectations.

1.3 Scope and standards

The project follows four standards from the start: plain-language explanations, strict honesty about limits, no claim of clinical validity, and pre-registration (rules fixed before running the model). Everything described here is reproducible from the project repository; Section A lists the main files.

2 Background

2.1 What a chest X-ray shows

A chest X-ray is a shadow picture. X-rays pass easily through air and are stopped by dense tissue, so air-filled lungs appear dark and bone, fluid and the heart appear white. The standard frontal view is the posterior-anterior (PA) film. By convention the patient's right appears on the left of the image, which matters later for left/right questions.

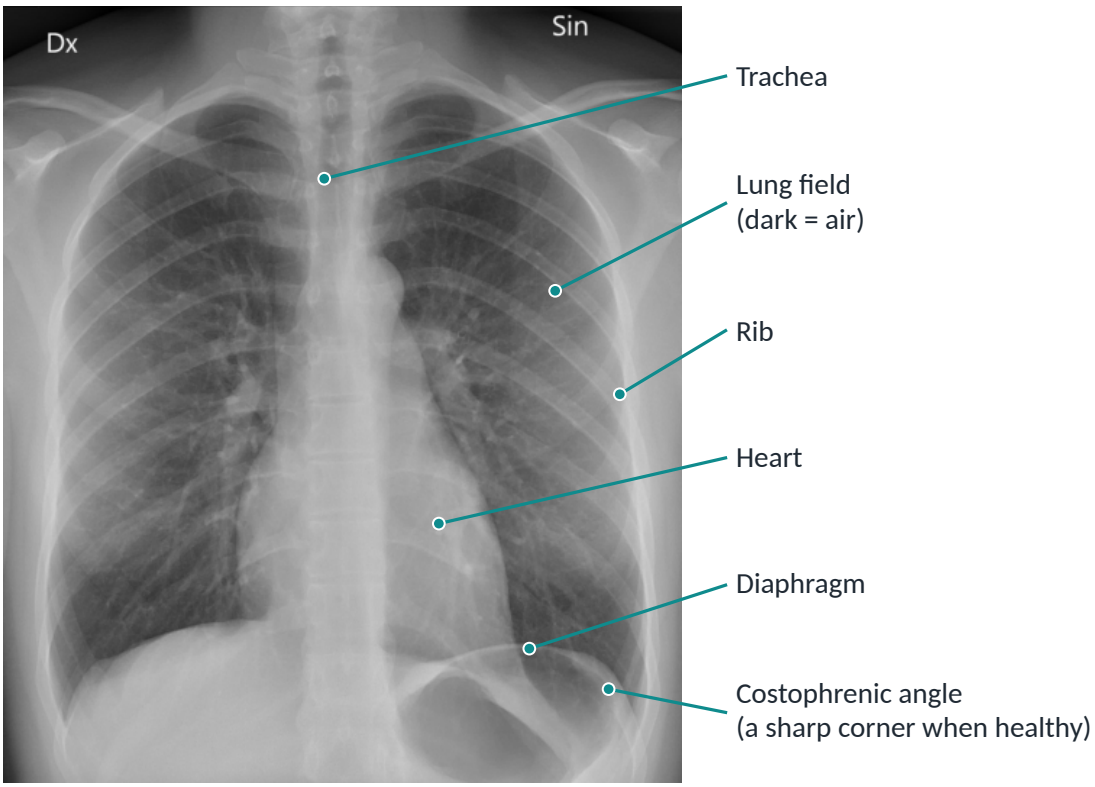


Figure 1. A normal frontal (PA) chest X-ray with the main structures labelled. "Dx" and "Sin" are Italian for right and left. Photograph: M. Häggström, Wikimedia Commons, CC0 [11].

Four findings recur throughout the project (Figure 1):

- **Cardiomegaly**: an enlarged heart, usually judged by a cardiothoracic ratio above 0.5 (heart width over chest width).
- **Pleural effusion**: fluid around the lung, which blunts the sharp angle where the diaphragm meets the chest wall.
- **Pneumothorax**: air outside the lung, seen as a dark area with no lung markings and a visible pleural line.
- **Atelectasis**: partly collapsed lung, which looks whiter and pulls nearby structures toward it.

2.2 Medical visual question answering

In medical VQA the model receives an image and a question and must write an answer (Figure 2). We separate two kinds of difficulty, because they fail in different ways: **perception** (seeing the finding in the picture) and **knowledge** (knowing facts about the disease, such as its cause or treatment).

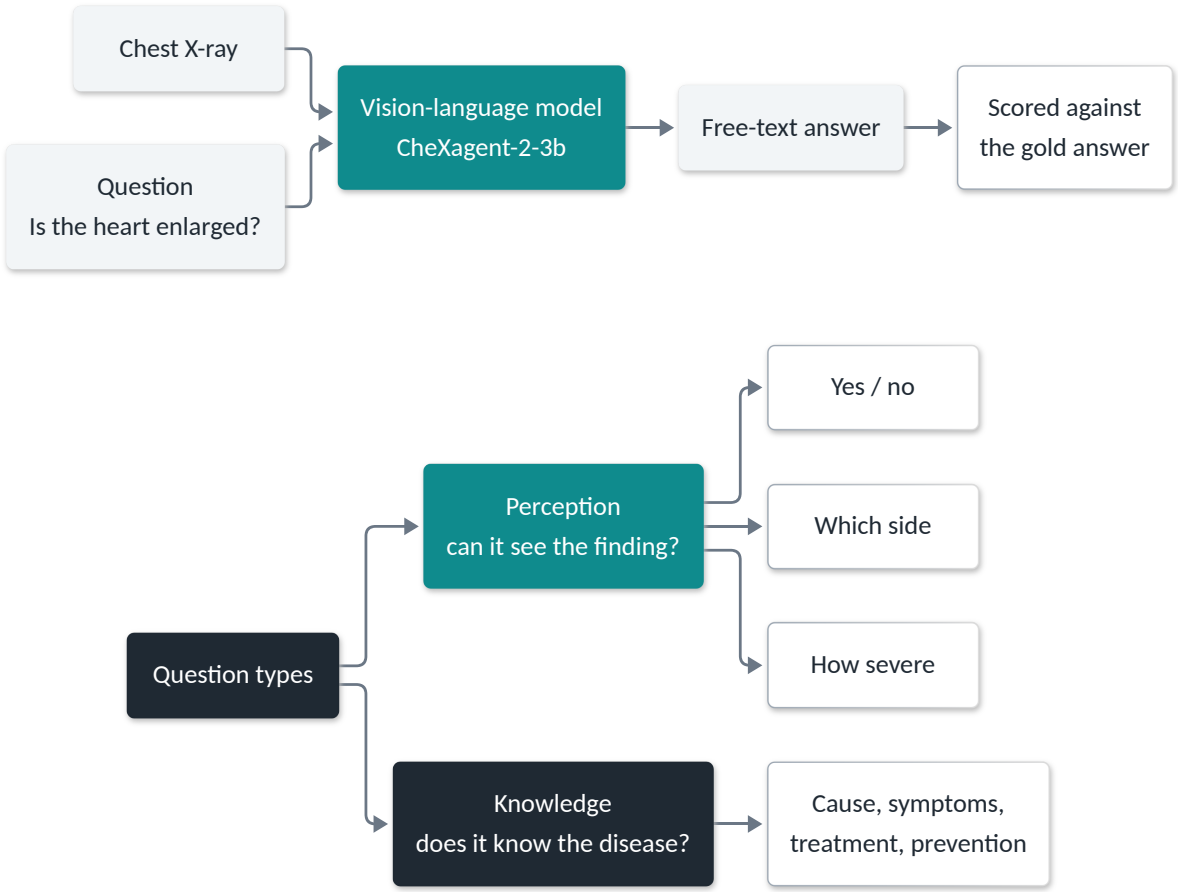


Figure 2. The task. Yes/no, side and severity questions test perception; knowledge questions test what the model knows about a disease.

2.3 Datasets and model

SLAKE [7] and **VQA-RAD** [8] are radiology VQA datasets with expert-written questions. We use their chest X-ray questions only. SLAKE additionally gives each image a set of labels, which we turn into a disease label for 130 chest images across eight classes (cardiomegaly 29, atelectasis 21, mass 19, nodule 18, pneumonia 17, effusion 13, pneumothorax 10, infiltration 9). The model is **CheXagent-2-3b** [1], a 3.1-billion-parameter vision-language model trained on chest X-ray instruction data. Its training data (CheXinstruct) already includes SLAKE and VQA-RAD, so our tests are close to in-distribution rather than a clean held-out test.

3 Literature review

We reviewed six papers and keep three here because they changed the plan; the other three gave useful context.

3.1 Papers that shaped the project

CheXagent [1] is the only model in our review built end to end for chest X-rays. It was trained on CheXinstruct, a collection of 8.5 million instruction triplets from 32 datasets covering 35

tasks, and outperforms larger general-purpose medical models. Its own results show where it is weakest: fine-grained reasoning such as side and severity reaches only 77–79%, against over 90% on coarse tasks. We chose it as the baseline model, and treat left/right and heart-size questions as the hardest perception cases.

RULE [2] adds retrieval to a medical vision-language model and shows why naive retrieval backfires. Too few or too many retrieved passages hurt, and the model over-trusts retrieved text: among errors introduced by retrieval, 47.4%, 47.4% and 58.7% (IU-Xray, FairVLMed, MIMIC-CXR) were cases where the model had answered correctly without it. RULE retrieves similar case reports, not authoritative knowledge. For us this means: keep retrieved text short, check that retrieval does no harm, and include an arm that supplies only the disease name without any retrieved text.

Med-Flamingo [3] evaluates answers with clinicians and finds that automatic scores can disagree with clinical judgement. We therefore plan a blind 0/1/2 rating of answers and avoid relying on a single automatic score.

3.2 Context papers

PubMedCLIP [4] found that the image encoder matters far more than the text encoder for medical VQA. **LLaVA-Med** [5] shows cheap curriculum tuning of a general model on GPT-4-written instruction data, which carries a hallucination risk because the writer never saw the image. **PMC-VQA** [6] builds questions from figure captions, and even with filtering such questions can often be answered from text alone, a warning that we later met directly (Section 5.4).

	Chest X-ray focus	Free-text answers	Uses retrieval	Clinician / blind rating	Textbook source	what we took from it
CheXagent (2024)	●	●	○	●	○	our baseline model; weakest on left/right and severity (77-79%)
RULE (2024)	○	●	●	○	○	naive retrieval hurts: 47-59% of its errors follow the text
Med-Flamingo (2023)	○	●	○	●	○	automated scores disagree with clinicians: rate blind
LLaVA-Med (2023)	○	●	○	○	○	cheap tuning; GPT-4-written training data
PMC-VQA (2023)	○	●	○	○	○	caption-made questions can be answered from text alone
PubMedCLIP (2023)	○	○	○	○	○	the image encoder matters more than the text encoder
This project	●	●	●	○	●	**none of the six papers combines these (hollow = planned)**

Figure 3. Comparison of the six papers and this project. Highlighted rows shaped the plan; the hollow circle marks the blind rating that is planned but not yet done.

3.3 Gap and design decisions

Among the papers reviewed, none combines a chest-X-ray model with retrieval of authoritative knowledge and a careful, pre-registered evaluation. Figure 4 shows how each finding became a concrete decision.

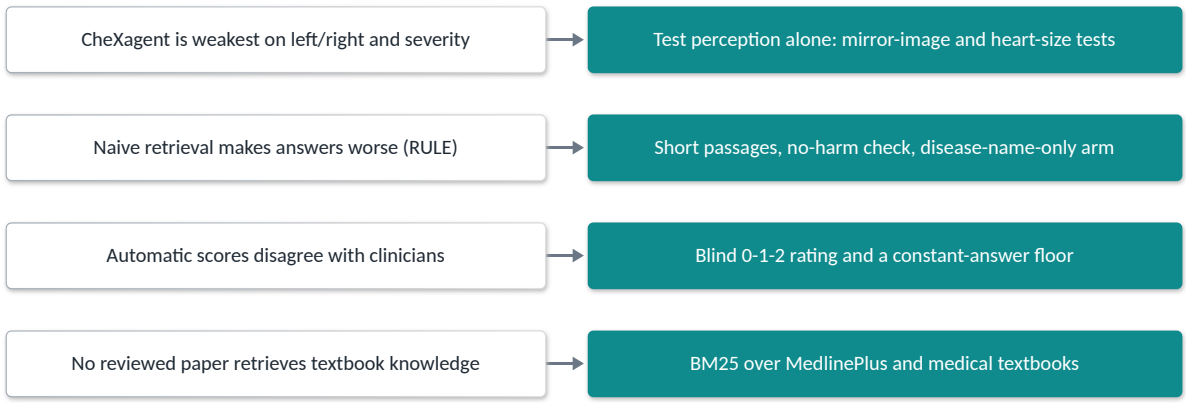


Figure 4. From literature findings to design decisions.

4 Evaluation method

4.1 Pre-registration

Before any run we wrote the rules into `docs/evaluation_protocol.md`: the test sets, the scorer, the statistics and the conditions that would count as success. Changes are recorded as dated amendments (six so far) rather than silent edits. Figure 5 shows the flow.

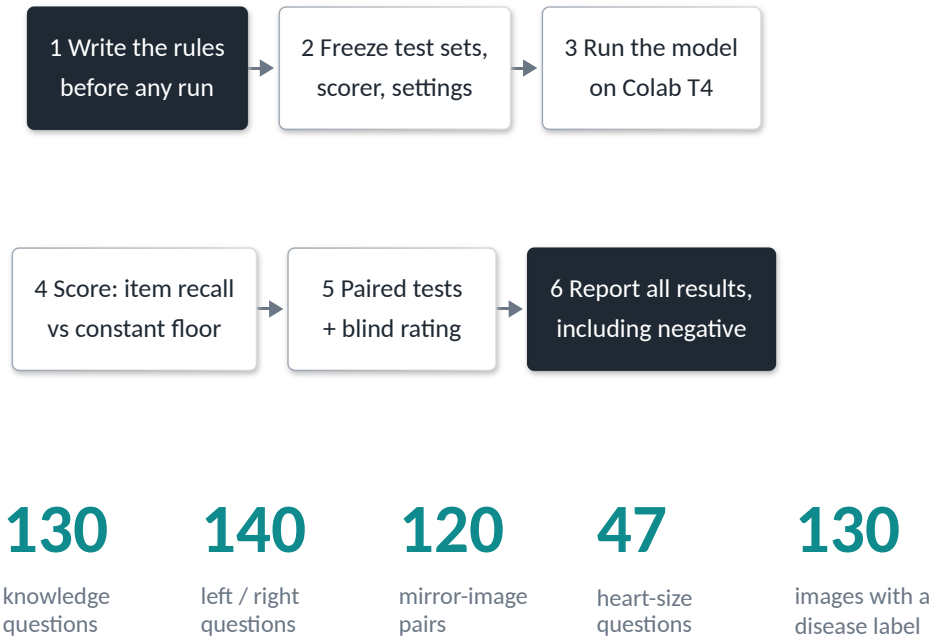


Figure 5. Evaluation flow and the test sets used so far.

4.2 Scoring

- **Exact match and loose match** for short answers (yes/no, one word).
- **Item recall** for list-style answers: the share of the gold answer's items that the model's answer covers (scorer version 2, frozen).

- **Constant-answer floor**: the score of a system that ignores the image and always writes the most common gold answer for that question type (leave-one-out). A real system must beat this floor.
- **Statistics**: bootstrap resampled by image, because several questions share one image; exact McNemar test for paired yes/no outcomes; exact sign test for paired scores.

4.3 Test sets

Set	Size	Purpose
Pilot	60	First feasibility run
Baseline (run 2)	219	General, left/right and severity questions
Knowledge (run 3)	130	Symptoms, treatment, cause, prevention, function, why
Side (run 3)	140	Questions whose answer is a side
Mirror pairs (run 4)	120	Same question on an image and its mirror
Heart-size (run 4)	47	"Is the heart enlarged?" wordings
Disease labels (run 5)	130	Naming the disease from the image

Table 1. Test sets. Five questions were removed from the original 224-question baseline after a keyword-filter bug admitted non-chest images (Section 5.1).

5 Experiments and results to date

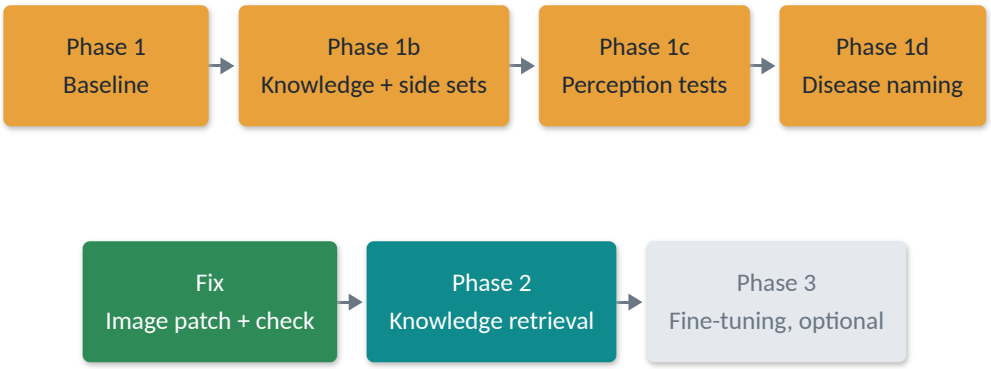


Figure 6. Project phases and their status. Amber phases are runs in which the image did not reach the model; green is the fix; teal is built and waiting for a model run.

Important reading note. Sections 5.1 to 5.3 report runs in which the model answered from the question text alone (see Section 5.4). They are valid as a *text-only control*: they show how much of these benchmarks can be solved without the image. They do **not** show what the model can or cannot read from an X-ray. Conclusions about perception are withdrawn until the runs are repeated with the fix.

5.1 Zero-shot baseline

The first pilot on 60 questions scored 37 of 60 (61.7%). The larger run (219 questions after removing five rows that were probably not chest images) is summarised in Table 2.

Question type	Correct / total	Accuracy	95% interval
Overall	128 / 219	58.4%	52–65%
General	70 / 96	72.9%	63–81%
Left/right	44 / 100	44.0%	35–54%
Severity	14 / 23	60.9%	41–78%

Table 2. Run 2 (text-only control). Intervals are Wilson 95%.

On the 22 heart-size yes/no questions the model answered "no" every time, although the gold answer was "yes" in eight. Left/right accuracy was lower than general accuracy, but the comparison is partly confounded by dataset and question style, so we treat it as likely rather than settled.

5.2 Knowledge questions

On 105 knowledge questions (excluding 25 "function" questions whose gold answer is always "Breathe") the model reached an item recall of 0.37, against 0.38 for the constant-answer floor (Figure 7, middle). It was above the floor only on symptoms (0.42 against 0.21), matched it on treatment and cause, and fell below it on prevention, where it wrote "keeping healthy" while the gold answers were mostly different phrasings. The 130 knowledge questions have only 33 distinct gold answers, so the answer is largely determined by which disease is in the image; this observation motivates the two-step design in Section 6.

5.3 Perception diagnostics

Two diagnostics were run on left/right behaviour. In the **mirror test**, each question was asked on the original image and on its horizontal mirror. A model that reads laterality should swap its answer. Of 73 pairs in which a side was named on both images, only 2 swapped (3%), and 112 of 120 answers were byte-identical; on the 29 explicit "which side" questions, 29 of 29 were identical and 26 of 29 answers were "Right". In the **wording test**, six prompt wordings were tried on 60 side questions; none fixed the behaviour. A heart-size wording test found a weak signal (sensitivity 9 of 22, specificity 22 of 25) below the bar fixed in advance.

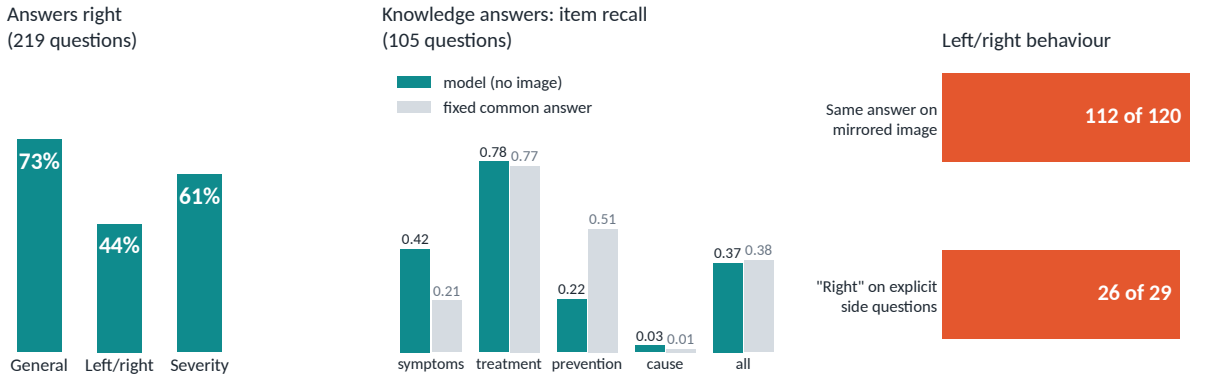


Figure 7. Text-only control results. Left: accuracy by question type with 95% intervals (219 questions). Middle: knowledge item recall against the constant-answer floor (105 questions). Right: left/right behaviour (mirror test).

5.4 Disease naming and the turning point

Phase 1d asked the model to name the disease on 130 labelled images in three ways (open question, forced choice, one-versus-rest scoring per class). All 130 images received the same answer under each method, per-class scores varied by only 0.002–0.009, and accuracy was 13% (open), 0% (forced) and 7% (per-class), against a majority-class rate of 22%. Identical answers on every image are not a perception failure; they are what one would expect if the model could not see the image.

Cause. With `transformers` 4.46.3 the generation routine passes an *empty* cache object to the model. CheXagent's own code encodes the image only when that cache is `None`, so the image branch was skipped silently and every answer came from the text of the prompt alone (Figure 8).

Fix and safeguard. We patched the condition so that an empty cache still encodes the image, and added an **IMAGE CHECK** to every notebook: the same file path is used with black, white and noise pixels, and the run stops if the output log-probabilities do not change by at least 0.01.

Consequences. All earlier documents now carry a correction banner (Amendment 6 of the protocol, experiment log entry 17). The earlier runs remain useful as a text-only control: 58% overall accuracy, with no image at all, shows how much of these benchmarks can be answered from the question wording alone, which is the same shortcut risk that PMC-VQA warns about. Conclusions about left/right and heart-size perception are withdrawn until the reruns (notebooks 04 and 05, then 01–03).

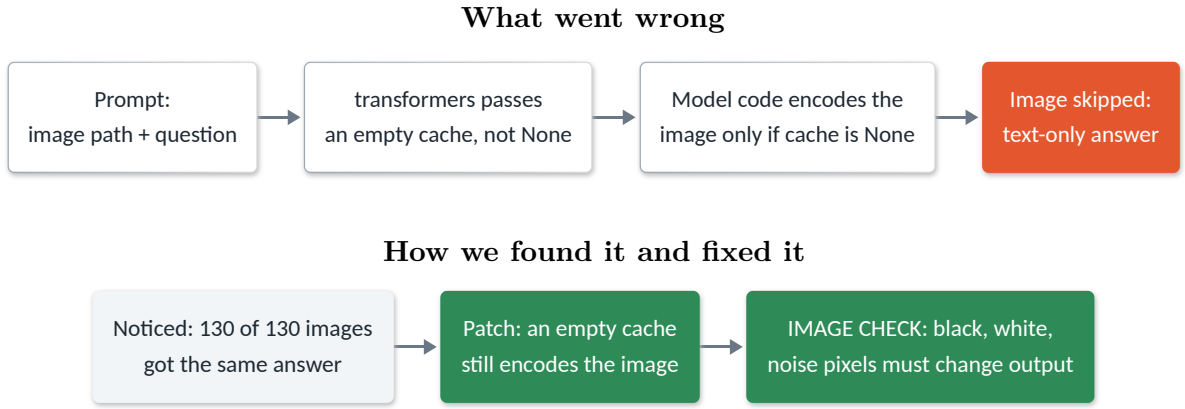


Figure 8. The image-skipping fault, how it was noticed, and the fix.

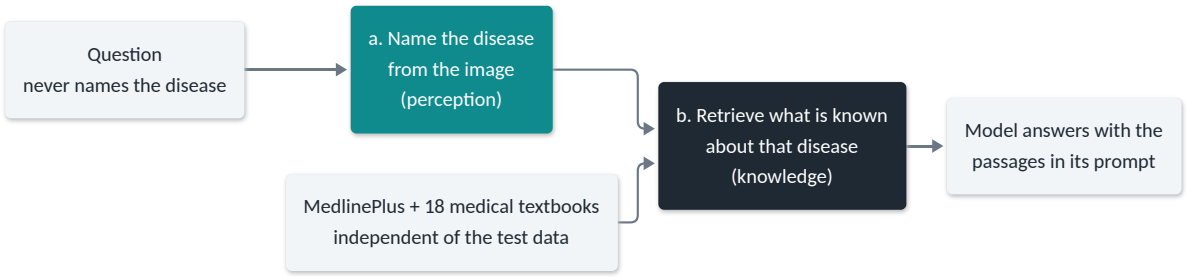
6 Phase 2: knowledge-grounded retrieval

6.1 Design

SLAKE's knowledge questions never name the disease ("What symptoms will the disease on the left bring?"), and the gold answer is effectively a lookup keyed by the disease on the image. The system therefore works in two steps (Figure 9): (a) name the disease from the image, a perception step; (b) retrieve what is known about that disease and let the model answer with the passages in its prompt, a knowledge step.

Retrieval uses our own BM25 implementation over two corpora that are independent of the test data: **MedlinePlus** (US National Library of Medicine, public domain) and the 18 medical

textbooks of the MedRAG collection [9]. The textbooks are copyrighted, so they are downloaded at run time into a git-ignored folder and never committed. The retrieval setting (topic lookup plus textbooks, two passages, 700 characters each) was chosen on 16 development questions only and frozen before held-out numbers were computed.



Disease on the image	Question about	Gold answer, on every such image
Pneumonia	cause	Bacterial infection
Pneumonia	treatment	Medical therapy, supportive therapy
Cardiomegaly	cause	Hypertension, dilated cardiomyopathy

Figure 9. The two-step pipeline, and why it is needed: the gold answer depends only on the disease on the image.

6.2 Experimental arms

Six arms (Figure 10) separate the value of knowing the disease from the value of retrieval. Oracle arms (A2–A4) use SLAKE's own disease label, which a real system would not have, so they are upper bounds and never reported as the result. The real system is A6. Amendment 5 requires A6 to beat both the fair baseline A1 and the guess-only arm A5, so a gain that comes only from naming the disease cannot be credited to retrieval.

Arm	Disease named	Retrieved text	Role
A1	none	none	fair baseline, same answer style
A2	none	from the question alone	oracle subset: why the disease matters
A3	true disease (oracle)	none	upper bound: value of knowing the disease
A4	true disease (oracle)	yes	upper bound: value of retrieval
A5	model's own guess	none	real: the guess alone
A6	model's own guess	yes	the system

oracle: uses SLAKE's own label, an upper bound, never the result real: uses only what the model outputs

Figure 10. The six experimental arms.

6.3 Retrieval check without a model

Before spending any GPU time we measured how often the retrieved text contains the gold answer's items (Table 3, Figure 11). "Strict" coverage requires the gold item's words to appear within one sentence.

Retrieval query	Dev (n=16)	Held-out (n=34)	All 50
True disease	0.42	0.29	0.33
Wrong disease	0.16	0.17	0.13
Question only	0.05	0.03	0.04

Table 3. Strict gold-item coverage of the retrieved text. The setting was tuned on the development questions only.

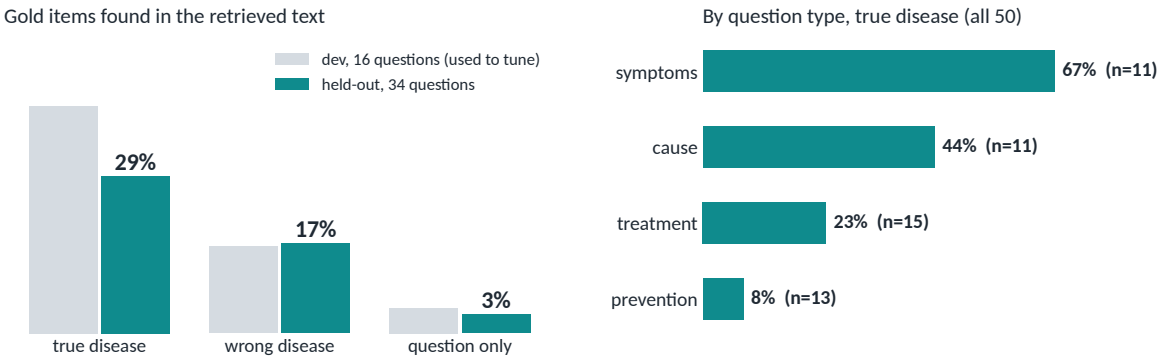


Figure 11. Left: coverage by query type. Right: coverage by question type with the true disease (all 50 questions).

Three readings follow. First, the corpus holds more of the answer when the disease is right, and almost none when only the question is used, which supports the two-step design. Second, coverage is an upper limit on what retrieval can add: even the best case covers about a third of the gold items. Third, it differs by question type (symptoms 67%, cause 44%, treatment 23%, prevention 8%), because generic gold phrases such as "keep healthy" almost never match a source sentence. A router that sends only knowledge questions to retrieval (25 of 219 baseline questions routed, all in the knowledge set) was also checked, to ensure retrieval cannot affect perception questions.

7 Status, limitations and next steps

7.1 Where the work stands

Built and tested: the baseline pipeline, all test sets, the scorers and analysis scripts, the retrieval pipeline and corpora loaders, the disease-naming and retrieval notebooks, and 121 automated tests (using stand-in models to check the plumbing). Not yet done: any model run with the image visible.

7.2 Next steps

1. Rerun the disease-naming notebook (04) first with the IMAGE CHECK, then in full, to see whether the model can name the disease at all when it sees the image. This sets the ceiling for the whole retrieval system.
2. Run the retrieval notebook (05) with the disease method chosen in step 1.
3. Rerun notebooks 01–03 to replace the text-only control with real perception measurements.
4. Collect blind 0/1/2 ratings of a sample of answers.
5. Consider targeted fine-tuning (Phase 3) only if Phase 2 shows a gap worth closing.

7.3 Limitations

- Zero-shot only, one model, and small test sets (the retrieval check has 50 questions).
- SLAKE's knowledge answers are annotator-written templates, so a high score does not imply clinical knowledge.
- CheXagent's training data includes SLAKE and VQA-RAD, so the tests are not a clean held-out evaluation.
- The disease labels come from SLAKE annotations and cover eight classes only.
- No clinician has reviewed any output. Nothing in this report supports a claim of clinical validity or safety.

8 Conclusion

The literature review gave a clear, narrow question: can retrieved textbook knowledge improve a chest-X-ray model's answers to knowledge questions, without making perception questions worse? We built an evaluation that fixes the rules in advance and a retrieval pipeline that measurably contains the answer more often when the disease is known. The most important result so far is a negative and instructive one: in every run, the image did not reach the model. Because of the safeguards we had set up (a constant-answer floor, identical-answer checks, pre-registered readings), the fault was found and traced to a software-version incompatibility, and the earlier numbers were relabelled as a text-only control instead of being reported as perception results. The next stage is a clean set of runs with the image visible, starting with disease naming.

References

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[10] S. Robertson, H. Zaragoza. *The Probabilistic Relevance Framework: BM25 and Beyond.* Foundations and Trends in Information Retrieval 3(4), 2009.

[11] M. Häggström. *Normal posteroanterior (PA) chest radiograph (X-ray).* Wikimedia Commons, CC0.

A Reproducibility and repository guide

Path	Content
`docs/evaluation_protocol.md`	Pre-registered rules and amendments 1–6
`docs/phase2_rag_plan.md`	Phase 2 design, arms, frozen settings
`docs/experiment_log.md`	Dated log of decisions and findings (entries 1–17)
`docs/literature_review.md`	Full review of the six papers
`baseline/`	Model runner, scorers, labels, retrieval code
`notebooks/01-05`	Colab notebooks (baseline, knowledge and side, diagnostics, disease naming, retrieval)
`results/`	Raw outputs of every run, including the archived text-only disease-naming run
`deliverables/diagrams/`	Mermaid sources of the block diagrams (`scripts/render_diagrams.py`)
`scripts/make_figures.py`	Regenerates the chart and photo figures

Table 4. Where to find things.

B Glossary

Atelectasis
Partial collapse of the lung.

BM25
A classic keyword-based ranking function for finding relevant text passages.

Cardiomegaly
An enlarged heart.

Constant-answer floor
Score of a system that ignores the image and always gives the most common answer.

Item recall
Share of the gold answer's items that an answer covers.

Oracle
An arm that uses the true label; an upper bound, not a real system.

Perception / knowledge
Seeing a finding in the image / knowing facts about the disease.

Pre-registration
Writing down test sets, scoring and success conditions before running the model.

RAG
Retrieval-augmented generation: adding retrieved text to the model's prompt.